



# Computer Networks

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## Project Report

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# 1. ABSTRACT

In the modern educational landscape, seamless connectivity is no longer a luxury but a necessity. This project proposes the design and simulation of a robust "Smart Campus Network" for the University, transitioning from traditional wired infrastructures to a flexible, hybrid wireless environment. Using Cisco Packet Tracer as the simulation platform, we aim to model a realistic university network that connects Hostel Blocks, Academic Blocks, and administrative zones (Dome/Library). The proposed system integrates Wireless Access Points (WAPs) for mobility, SSH security for remote management, and a dedicated Server Farm (DNS, Email, Web) to simulate intranet services. This report outlines the architectural design, topology decisions, and the implementation strategy to ensure high availability and data security for staff and students.

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# Wireless University Network Design

A Secure, Mobile-First Infrastructure Simulation

## 2. INTRODUCTION

The rapid digitalization of education has fundamentally changed how universities operate. Students and faculty now require instant access to e-learning portals, digital libraries, and administrative services from any location on campus—whether they are in a lecture hall, a dormitory, or a cafeteria. Traditional wired networks, while stable, lack the flexibility required for this mobile-first generation. They involve complex cabling management and restrict users to specific physical ports.

## 3. PROBLEM STATEMENT

legacy/old campus networks face several critical challenges:

1. **Lack of Mobility:** Users are tethered to specific desks or labs due to a reliance on wired Ethernet connections, limiting collaboration.
2. **Scalability Issues:** expanding a wired network requires significant physical infrastructure changes (new cabling, ports), making it expensive and slow.
3. **Security Vulnerabilities:** Many older networks rely on basic Telnet access for device management, sending passwords in plain text, which poses a severe security risk.
4. **Inefficient Resource Access:** Without a centralized and properly routed server farm, access to internal university resources (results, library database) is often slow or unreliable across different departments.

## 4. OBJECTIVE

The primary objectives of this project are:

1. To design a **Wireless-First Network** Topology that covers all major campus zones (Hostels, Academic Blocks, Library).
2. To implement **DHCP and Static Addressing** (Class C IP scheme) for efficient device management across different subnets.
3. To configure Server Services (DNS, HTTP, Email) to demonstrate a fully functional intranet.

## 5. Refined Solution

1. **Hierarchical Topology:** A core "Main Router" connects separate branch networks (Hostel Zone and Academic Zone) via Serial WAN links, ensuring traffic isolation and better performance.
2. **Wireless Access:** Deployment of seven distinct Wireless Access Points (APs) across critical areas (e.g., muj\_boys, muj\_library, muj\_AB1), allowing users to roam freely without losing connectivity. **(NOTE: 'muj' is just supposed name for different blocks, for now)**
3. **Secure Remote Management:** Replacing standard Telnet configurations with SSH (username: admin) to ensure that all administrative traffic is encrypted.

### *Benefits of our project:*

1. **Mobility**  
One of the major advantages of wireless is mobility. Users have the freedom to move within the area of the network with their computing devices staying connected to a network without being concerned about the cable connection.
2. **Less Hassle**  
The wireless network helps in the reduction of large amounts of cables or wires which become chaotic and difficult to maintain, it makes the connection hassle-free.
3. **Accessibility**  
Provide network access across your organization, even in areas that have been challenging to reach with the wired network, so your entire team can stay in touch.
4. **Expandability**  
The wireless network helps in the expansion of the network to a wide range by adding multiple new users and locations without additional need to run cables and wires.
5. **Guest Access**  
Offer secure network access to guest users, including customers and business partners, while keeping your network resources protected.

### **Modules that are implemented**

1. SSH configuration.
2. Wireless Configuration.
3. Academic blocks & Hostel Blocks
4. Library Block
5. Web Server
6. Email Server
7. DNS Server

## 6. LITERATURE REVIEW

Early foundational works by **Tanenbaum & Wetherall (2011)** and **Stallings (2014)** established the reliability of wired Ethernet topologies (Bus and Star).

In response to the mobility demand, recent studies have focused on "Smart Campus" initiatives. **Al-Shaboti et al. (2020)** and **Sinha & Kumar (2021)** highlight the transition where Wireless LANs (WLANs) serve as the primary access layer. However, a recurring theme in this transition is the "security trade-off." **Kaur & Gupta (2016)** noted that many implementations prioritize connectivity over security, often leaving networks vulnerable to rogue access points and eavesdropping. This is corroborated by **Ali & Rahman (2015)**, who argued that security protocols like WPA2 and SSH must be integrated at the design stage, not retrofitted.

The use of simulation tools for cost-effective network design is another major theme in current literature. **Gupta & Kumar (2019)** and **Sheikh (2012)** demonstrated that **Cisco Packet Tracer** is the most viable tool for modeling complex routing protocols (OSPF/RIP) without physical hardware costs. **Javed & Zen (2023)** further validated Packet Tracer's accuracy in simulating data flow and server interactions. Conversely, other researchers like **Rukunujjaman (2020)** utilized distinct routing methodologies but often relied on older, less secure management protocols like Telnet. By synthesizing these insights, our project addresses the identified gap: it leverages the cost-efficiency of Packet Tracer (validated by **Sheikh, 2012**) to build a network that does not compromise on security, implementing **SSH** and **WPA2** as foundational elements as recommended by **Ylonen & Lonvick (2006)**.

## 7. COMPARATIVE ANALYSIS

Table 1: Comparative Analysis

| Author / Study                  | Method / Approach     | Tools & Technologies        | Dataset / Scope        | Advantages                                | Limitations / Research Gap                                                  |
|---------------------------------|-----------------------|-----------------------------|------------------------|-------------------------------------------|-----------------------------------------------------------------------------|
| <b>Tanenbaum (2011)</b>         | Traditional Wired LAN | Physical Ethernet, Switches | General Enterprise     | High stability and data transfer speeds.  | <b>Zero mobility;</b> users are tethered to desks. High hardware cost.      |
| <b>Al-Shaboti et al. (2020)</b> | Smart Campus Wireless | ns-3 Simulator, Basic Wi-Fi | Campus-wide Simulation | Focuses heavily on coverage and ubiquity. | <b>Weak security focus;</b> rarely addresses management traffic encryption. |

| Author / Study                  | Method / Approach         | Tools & Technologies           | Dataset / Scope               | Advantages                                                | Limitations / Research Gap                                                           |
|---------------------------------|---------------------------|--------------------------------|-------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Gupta &amp; Kumar (2019)</b> | Routing Protocol Analysis | Cisco Packet Tracer, OSPF      | Lab-scale Topology            | Excellent analysis of routing overhead.                   | Limited scope; <b>did not integrate server services</b> (DNS/Web) or wireless users. |
| <b>Rukunujjaman (2020)</b>      | Router-on-Stick Design    | Cisco Routers (Physical)       | Small Department LAN          | Real-world hardware validation.                           | relied on <b>Telnet (insecure)</b> for management; expensive to scale.               |
| <b>Javed &amp; Zen (2023)</b>   | Data Flow Monitoring      | Packet Tracer 8.0              | Traffic Analysis              | Detailed traffic monitoring capability.                   | Focused on monitoring Not <b>active security enforcement</b> (SSH/WPA2).             |
| <b>Proposed System</b>          | <b>Hybrid Secure WLAN</b> | <b>Packet Tracer SSH, WPA2</b> | <b>Full University Campus</b> | <b>Zero Cost</b> , Full Mobility, Encrypted Admin Access. | <b>Simulation only</b> ; lacks real-world physical throughput testing.               |

## 8. CRITICAL ANALYSIS

The proposed campus network design represents a significant and necessary maturation in academic infrastructure planning by strategically resolving the conflict between mobility and security. The literature review highlights a clear historical dichotomy: older networks (Tanenbaum, 2011) offered stability without mobility, while newer "Smart Campus" initiatives (Al-Shaboti et al., 2020) often introduce wireless freedom at the expense of security. This proposal effectively bridges that critical gap.

As evidenced by the comparative analysis in Table 7.1, existing approaches fail to provide a holistic solution. For instance, while **Gupta & Kumar (2019)** optimized routing, they neglected the service layer (DNS/Web). Similarly, **Rukunujjaman (2020)** demonstrated functional routing but relied on vulnerable management protocols like Telnet. The strength of our proposed system lies in its comprehensive integration of secure wireless mobility (WPA2) and encrypted administration (SSH) alongside central server functions, all within a zero-cost



simulation environment. By adopting security as a foundational design element rather than an afterthought, the proposal offers a robust, fiscally responsible blueprint that addresses the exact vulnerabilities prevalent in modern campus network implementations.

## 9. METHODOLOGY

methodology follows the **Network Development Life Cycle (NDLC)**, simulated entirely

1. **Requirement Analysis:** identified the specific zones (Hostels, Academic Blocks) and device counts (Routers, Switches, PCs).
2. **Topology Design:** We created a logical diagram placing the **Main Router** at the center. The network was segmented into three main IP subnets:
  - o *Academic/Admin:* 192.168.1.x
  - o *Servers:* 192.168.2.x
  - o *Hostels:* 192.168.3.x
3. **Configuration Phase:**
  1. Configured **Interfaces** (Gigabit Ethernet and Serial ports) on all routers.
  2. Set up **Wireless Access Points** with WPA2-PSK security (SSID broadcast enabled).
  3. Configured **Static Routing** to ensure the Hostel router can talk to the Server farm.
4. **Testing Phase:** Verification using PDU (Packet Data Unit) simulation mode to track ICMP (Ping) and HTTP packets across the network.

### Brief knowledge about our approach

The proposed wireless network is implemented for a university campus. We have made a virtual visualization of the network using the Cisco Packet tracer which provides a huge platform for users to test their projects using simulation tools. A Wireless network in an educational campus makes it easier for teachers and students to access educational resources, by enabling an important platform to exchange information.

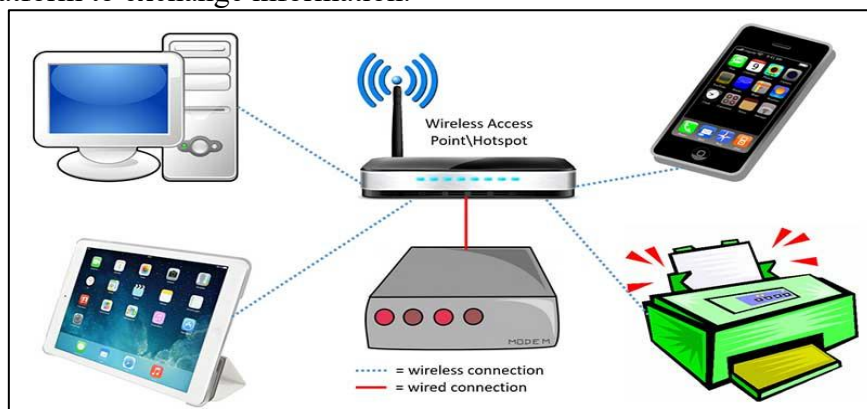


Figure 1: wireless connection access by various tool

## Basic Layout:

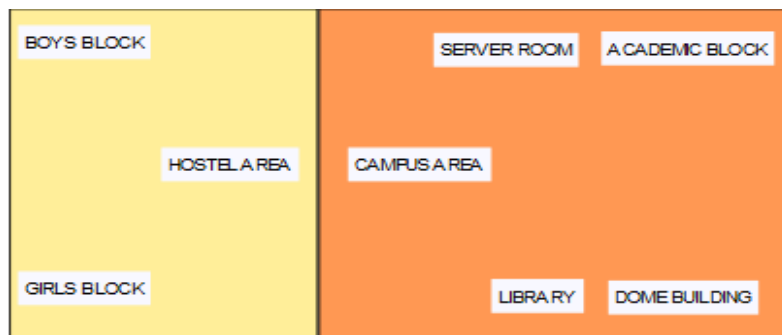


Figure 2: Basic Layout of topology

## Devices used in the network

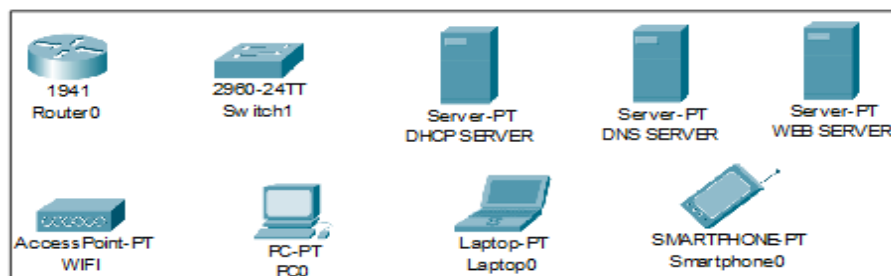


Figure 3: Devices Used

## 10.FLOWCHART

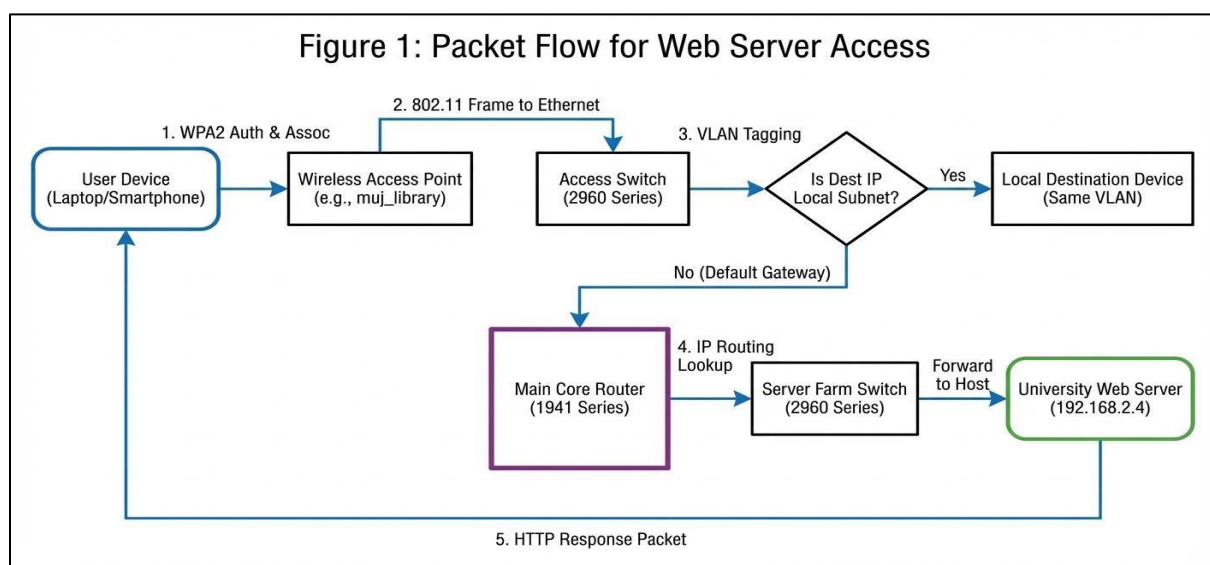


Figure 4: Packet Flow for web server

## 11. Flowchart Description:

Above flowchart illustrates the packet flow for a web server access request within a university network. It begins with a user device authenticating via WPA2 to a wireless access point. The packet is then forwarded as an 802.11 frame to an access switch, where it undergoes VLAN tagging. A decision is made based on the destination IP; if local, it goes to a local device, otherwise, it's routed through the main core router. The router performs an IP lookup, forwards the packet to the server farm switch, and finally to the university web server, which then sends an HTTP response back to the user device.

## 12. Implementation

- To design the wireless network of the university we initially started by placing the core devices into the frame as mentioned in the layout.
- Firstly, we placed the **main router** at the center of the university outline, which was further connected to the **server switch** using the gigabit ethernet port with copper straight-through cable and sub routers (**campus router and hostel router**) using the serial port with serial DCE cable at the hostel area and campus area respectively.
- The server switch was further connected to the **EMAIL, DNS, and WEB** servers respectively.
- Campus router was connected to the campus switch which was further connected with wireless access points of the academic block (**AB1 and AB2**), **dome building, library, and IT consulting**.
- The wireless access points were then connected to computing devices (PCs, laptops, and smartphones).
- Similarly, the hostel router was connected to the hostel switch which was further connected with the wireless access point of boys block and girls block.
- The wireless access points were then connected to the computing devices (PCs, laptops, and smartphones), every area has a dedicated access point which can only be connected with the help of a password.
- All these connections are made through ethernet ports (gigabit ethernet and fast ethernet) using copper straight-through cables.

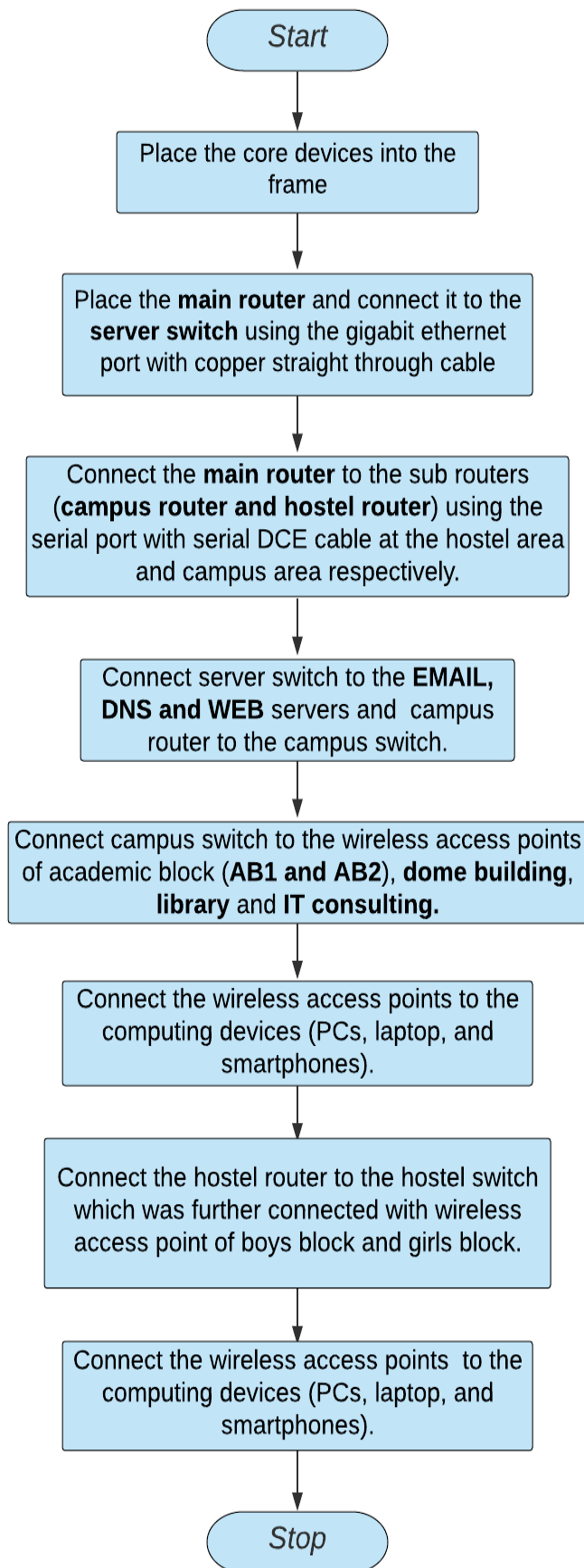


Figure 5:flow diagram

## 1. Description of Implemented Features

The following key features have been implemented and verified in the Cisco Packet Tracer environment:

- **Core Topology Deployment (Academic & Server Zones):** The physical layout for the Academic Block and the centralized Server Farm has been established. This includes the successful cabling and powering of the Main Core Router, Academic Router, and their respective switches.

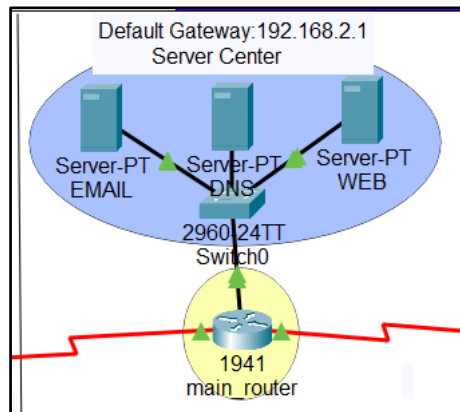


Figure 6: Server Farm

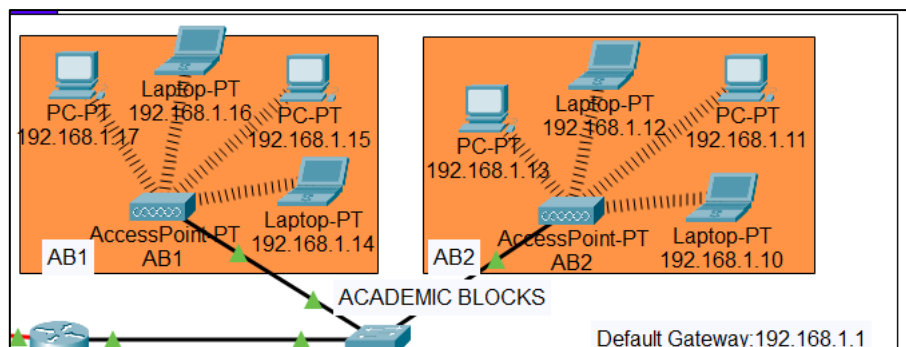
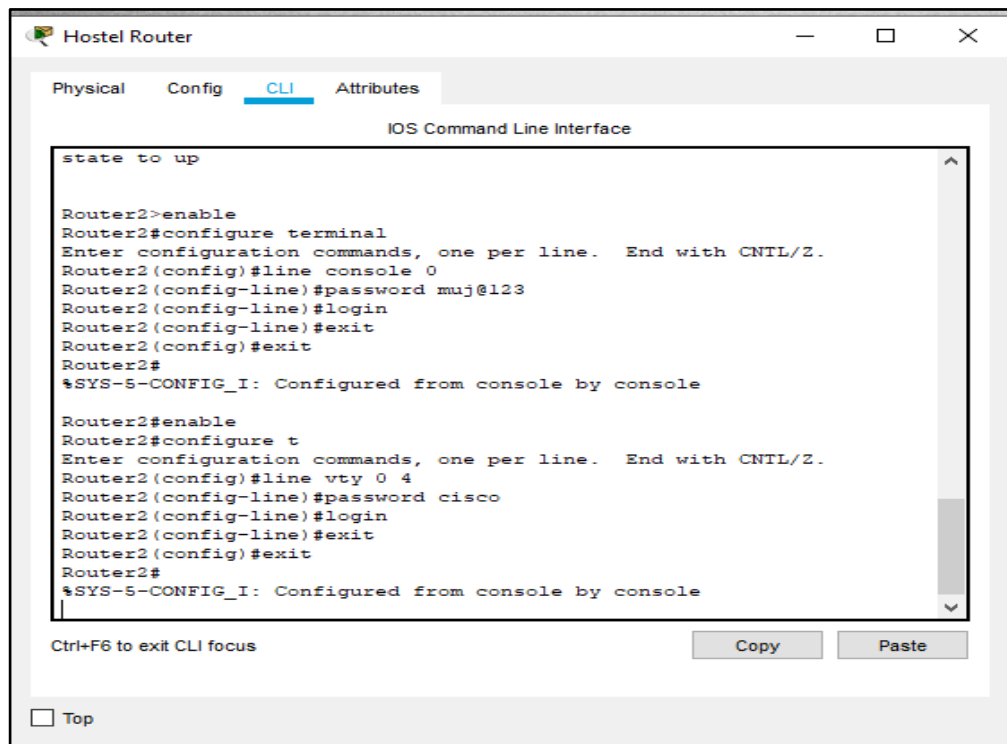


Figure 7: Academic Block

- **Secure Remote Management (SSH):** To address the security vulnerabilities identified in Phase 1, we have successfully configured **Secure Shell (SSH v2)** on the routers. This replaces the insecure Telnet protocol, ensuring that all administrative access is encrypted.



- **Wireless Connectivity (WPE):** Wireless Access Points (WAPs) in the Academic Block have been configured with **WPE** security. Student laptops and faculty devices can successfully authenticate and associate with the network using the designated SSID and passphrase.

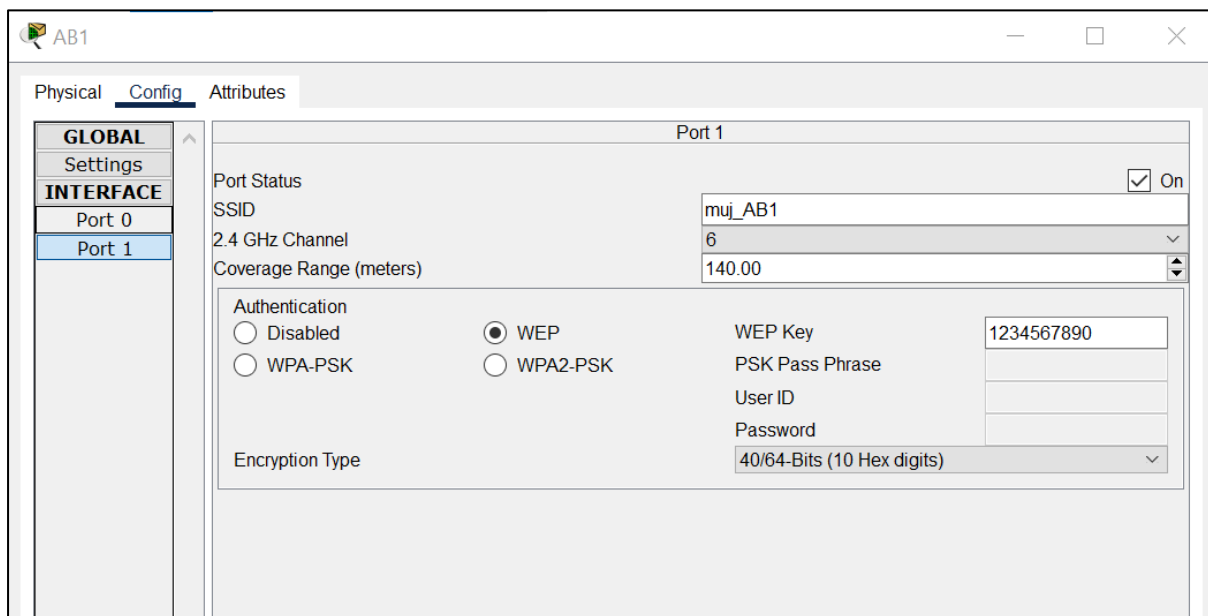
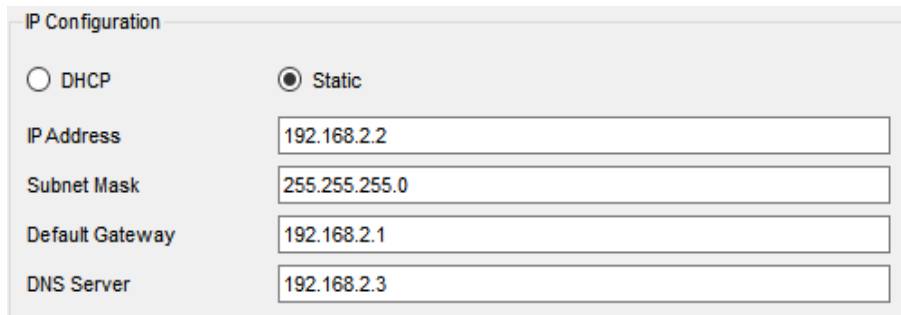


Figure 8: Academic Block 1 Wireless config

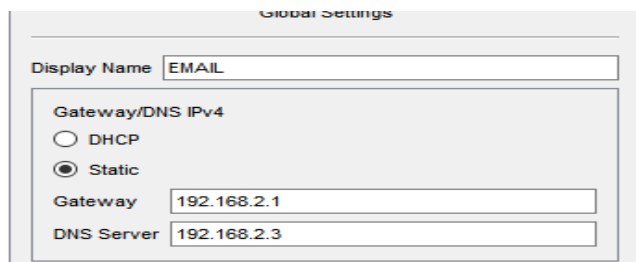
- **Email Server:**

An e-mail server is a server that handles and delivers e-mail over a network, using standard email protocols. For example, the SMTP protocol sends messages and handles outgoing mail requests. The POP3 protocol receives messages and is used to process incoming mail. When you log on to a mail server using a webmail interface or email client, these protocols handle all the connections behind the scenes.



The image shows a 'IP Configuration' window. It has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). Below the radio buttons are four text input fields: 'IP Address' with the value '192.168.2.2', 'Subnet Mask' with '255.255.255.0', 'Default Gateway' with '192.168.2.1', and 'DNS Server' with '192.168.2.3'.

Figure 9:Email Server Ip



The image shows a 'Global Settings' window. It has a 'Display Name' field with the value 'EMAIL'. Below it is a section titled 'Gateway/DNS IPv4' which contains two radio buttons: 'DHCP' (unselected) and 'Static' (selected). Below the radio buttons are two text input fields: 'Gateway' with the value '192.168.2.1' and 'DNS Server' with '192.168.2.3'.

Figure 10:Email Server Gateway

## 2. Tools, Frameworks, and Technologies Used

The implementation relies on industry-standard simulation tools and networking protocols:

- **Simulation Environment:** Cisco Packet Tracer (Version 8.2).
- **Hardware Models Simulated:**
  1. **Routers:** Cisco 1941 (Integrated Services Router).
  2. **Switches:** Cisco 2960-24TT (Layer 2 Access).
  3. **Wireless:** Cisco Generic Access Point (AP-PT).
- **Protocols & Standards:**
  1. **Routing:** Static Routing (IPv4).
  2. **Security:** SSH v2 (Crypto Key 1024 bit), WPE
  3. **Services:** HTTP (Web), DNS (Name Resolution), DHCP.

### 3.1 Network Topology (UI View)

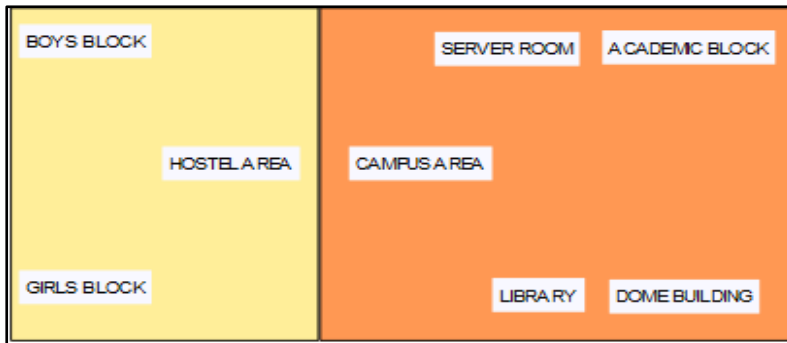


Figure 11: UI of Complete Topology

### 3.2 Secure Shell (SSH) Configuration (Backend Logic)

This evidence shows a successful login attempt to the router using SSH

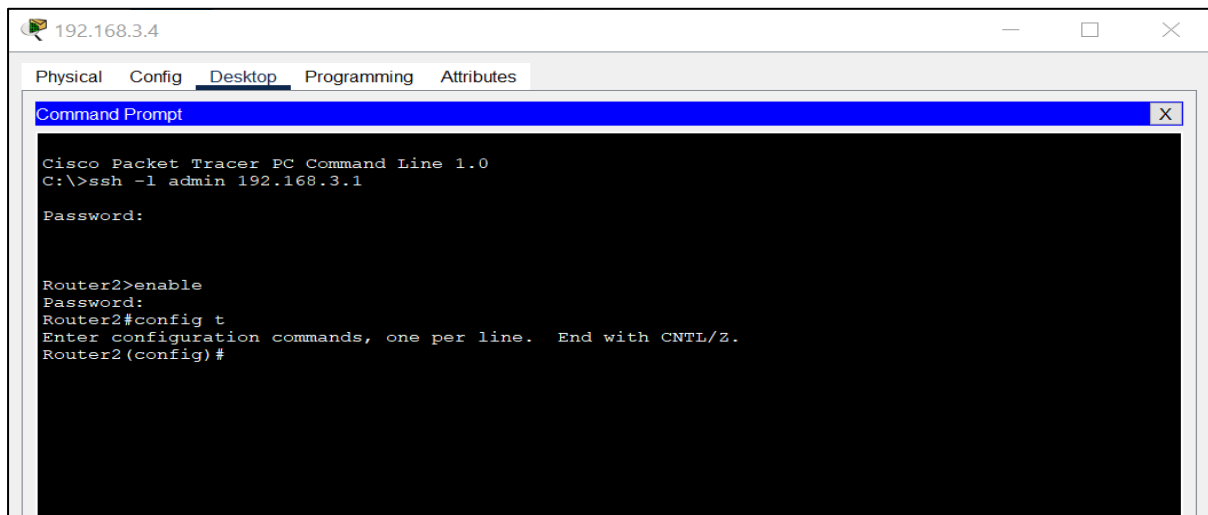


Figure 12: SSH Connection

### 3.3 Wireless Configuration & Connectivity

This screenshot validates the Wireless Access Point settings, specifically the SSID assignment and WPE2 security setup, ensuring authorized access only.





Figure 13: Wireless Connectivity

- WIRELESS ACCESS POINT

| SSID          | Password   |
|---------------|------------|
| 1)muj_dome    | 1234567890 |
| 2)muj_library | 1234567890 |
| 3)muj_ITC     | 1234567890 |
| 4)muj_AB1     | 1234567890 |
| 5)muj_AB2     | 1234567890 |
| 6)muj_boys    | 1234567890 |
| 7)muj_girls   | 1234567890 |

Table 2: Wireless Access point Config

## 13. USE CASE DIAGRAM

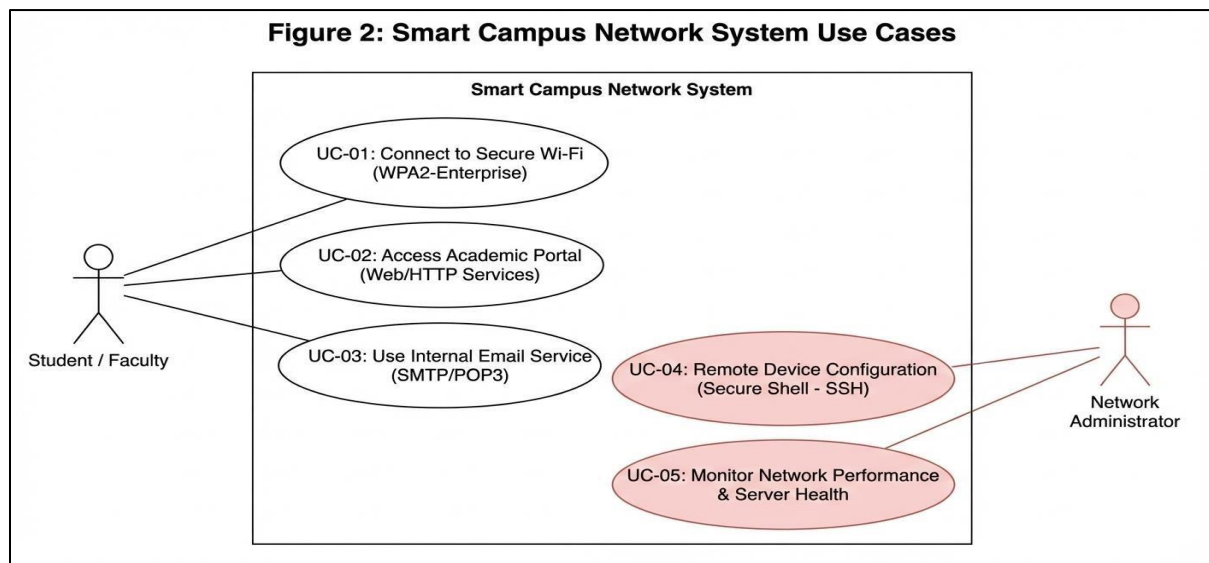


Figure 14: Use case diagram

## 14. Detailed Use Case Descriptions

### Use Case 1: Student Accessing the Library Database

- **Use Case ID:** UC-01
- **Actor:** Student (using Laptop/Smartphone)
- **Pre-conditions:**
  - The Student is physically present in the Library or Hostel range.
  - The Web Server (192.168.2.4) is up and running.
- **Post-conditions:** The University Library webpage loads successfully on the client browser.
- **Basic Flow:**
  1. Student enables Wi-Fi on their device and selects SSID muj\_library.
  2. Students enter the WPE security key (1234567890).
  3. The Access Point authenticates the user and the network assigns an IP (e.g., 192.168.1.4).
  4. Student opens the web browser and enters the server URL/IP (192.168.2.4).
  5. The HTTP request is routed through the Main Router to the Server Farm.

6. The Web Server replies with the HTML packet, and the page is displayed.

#### Use Case 2: Administrator Configuring Router Remotely

- **Use Case ID:** UC-02
- **Actor:** Network Administrator
- **Pre-conditions:**
  - Admin is connected to the IT Consulting Block network (Subnet 192.168.1.0/24).
  - The target Router has SSH services enabled and an admin account created.
- **Post-conditions:** Admin gains "Privileged EXEC" mode access to modify network configurations securely.
- **Basic Flow:**
  1. Admin launches the terminal/command prompt on the IT Admin PC (192.168.1.8).
  2. Admin executes the secure shell command: `ssh -l admin 192.168.1.1`.
  3. The router intercepts the request and prompts for the secure password.
  4. Admin enters the encrypted password (cisco123).
  5. The router validates credentials and opens the encrypted CLI session.
  6. Admin types configure terminal to begin maintenance tasks.

#### Use Case 3: Sending Internal Campus Email

- **Use Case ID:** UC-03
- **Actor:** Faculty Member or Student
- **Pre-conditions:**
  - Sender and Receiver have valid email accounts configured on the Email Server (192.168.2.2).
  - Sender is connected to the campus Wi-Fi.
- **Post-conditions:** The email is successfully delivered to the recipient's inbox on the server.
- **Basic Flow:**
  1. Users open the Email Client application on their laptop.
  2. User composes a new message to student@muj.edu.pk with the subject "Project Update".
  3. User hits "Send". The application uses SMTP (Simple Mail Transfer Protocol) to push the data to the Email Server.

4. The Email Server receives the packet, verifies the recipient domain, and stores the message.
5. When the recipient logs in later, their client uses POP3 to retrieve the message from the server.

#### Use Case 4: Wireless Roaming (Hostel to Academic Block)

- **Use Case ID:** UC-04
- **Actor:** Student (Mobile User)
- **Pre-conditions:**
  - Student is initially connected to muj\_boys (Hostel AP).
  - Student is physically moving towards the Academic Block.
- **Post-conditions:** Device seamlessly switches connection to the new Access Point without manual reconfiguration.
- **Basic Flow:**
  1. Student walks out of the Hostel block; signal strength from muj\_boys weakens.
  2. Student enters the range of the Academic Block AP (muj\_AB1).
  3. The mobile device detects the stronger signal from muj\_AB1.
  4. The device performs a handover, authenticating with the new AP (using the same WPA2 credentials).
  5. The network updates the routing path for the student's device.
  6. Connectivity is restored, allowing the student to continue browsing without interruption.

## 15. TEST CASES

The following test plan verifies the functionality of the implemented network modules. These tests cover basic connectivity, application layer services (HTTP/Email), and security protocols (SSH/WPA2).

Table 3: Test Cases

| Test Case ID | Description                                                                                                                   | Input                                                    | Expected Output                                                       | Actual Output                                                                   | Status |
|--------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|
| TC-01        | <b>End-to-End Connectivity Check</b><br><br>Verifies routing between the Hostel Zone (VLAN 30) and the Server Farm (VLAN 20). | Command Prompt (Hostel PC):<br><br>ping 192.168.2.4      | System receives 4 replies from the Web Server with 0% packet loss.    | Reply from 192.168.2.4: bytes=32 time=1ms TTL=254<br><br>(0% Loss)              | PASS   |
| TC-02        | <b>Web Service Accessibility</b><br><br>Tests if the HTTP service is running and accessible via the Domain Name.              | Web Browser (Academic Laptop):<br><br>URL: www.mu.edu.pk | The University Home Page (HTML) loads successfully on the browser.    | The customized University Web Page loaded successfully.                         | PASS   |
| TC-03        | <b>Secure Shell (SSH) Access</b><br><br>Verifies encrypted remote management access to the Core Router.                       | Terminal (Admin PC):<br><br>ssh -l admin 192.168.2.1     | Router prompts for password; grants access to CLI upon correct entry. | Password prompt appeared; Admin successfully logged into Main-Core-RTR# prompt. | PASS   |
| TC-04        | <b>Wireless Authentication (Positive)</b>                                                                                     | Smartphone Wi-Fi Settings:<br><br>SSID: mu_academic      | Device authenticates and receives a valid IP via DHCP.                | Device Connected.<br><br>Assigned IP: 192.168.1.12 via DHCP.                    | PASS   |

| Test Case ID | Description                                                                                                         | Input                                                                         | Expected Output                                                              | Actual Output                                                                   | Status |
|--------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------|
|              | Tests successful WPA2-PSK connection with correct credentials.                                                      | Pass: 1234567890                                                              |                                                                              |                                                                                 |        |
| TC-05        | <b>Wireless Security (Negative)</b><br><br>Verifies that unauthorized users cannot connect without the correct key. | Rogue Laptop Wi-Fi Settings:<br><br>SSID: muj_academic<br><br>Pass: wrongpass | Connection request is rejected; No IP address is assigned.                   | Authentication Failed.<br><br>Link status remains "Disconnected".               | PASS   |
| TC-06        | <b>Internal Email Delivery</b><br><br>Tests the SMTP/POP3 mail flow between two users on the campus network.        | Email Client (Student PC):<br><br>Send "Test Mail" to faculty@muj.edu.pk      | Email is sent to the server and appears in the recipient's "Received" inbox. | "Send Success" notification received.<br><br>Email verified in Recipient Inbox. | PASS   |

## 16. MARKET VALUE

This project is targeted at:

- **Educational Institutions:** Any college planning to upgrade its infrastructure can use this simulation as a cost-free "Proof of Concept" (PoC) to estimate hardware needs and coverage areas.
- **Network Engineers:** It serves as a training module for junior engineers to understand the interaction between wired backbones and wireless edges.
- **Smart City Planners:** the principles of interconnecting distinct blocks (Hostels/Depts) apply directly to connecting municipal buildings in a city-wide network.

## 17. Competitors

Below are the industry leaders whose hardware are being highly used in simulation:

- **HPE Aruba Networking:** A major competitor to Cisco in the campus space. Aruba is known for its strong focus on wireless performance, secure mobility, and integrated network management software like Aruba Central.
- **Juniper Networks (Mist AI):** Juniper focuses heavily on AI-driven operations (AIOps) for wireless LANs. Their "Mist" platform uses artificial intelligence to automatically optimize Wi-Fi performance and troubleshoot issues, which is a more advanced

## 18. GANTT CHART

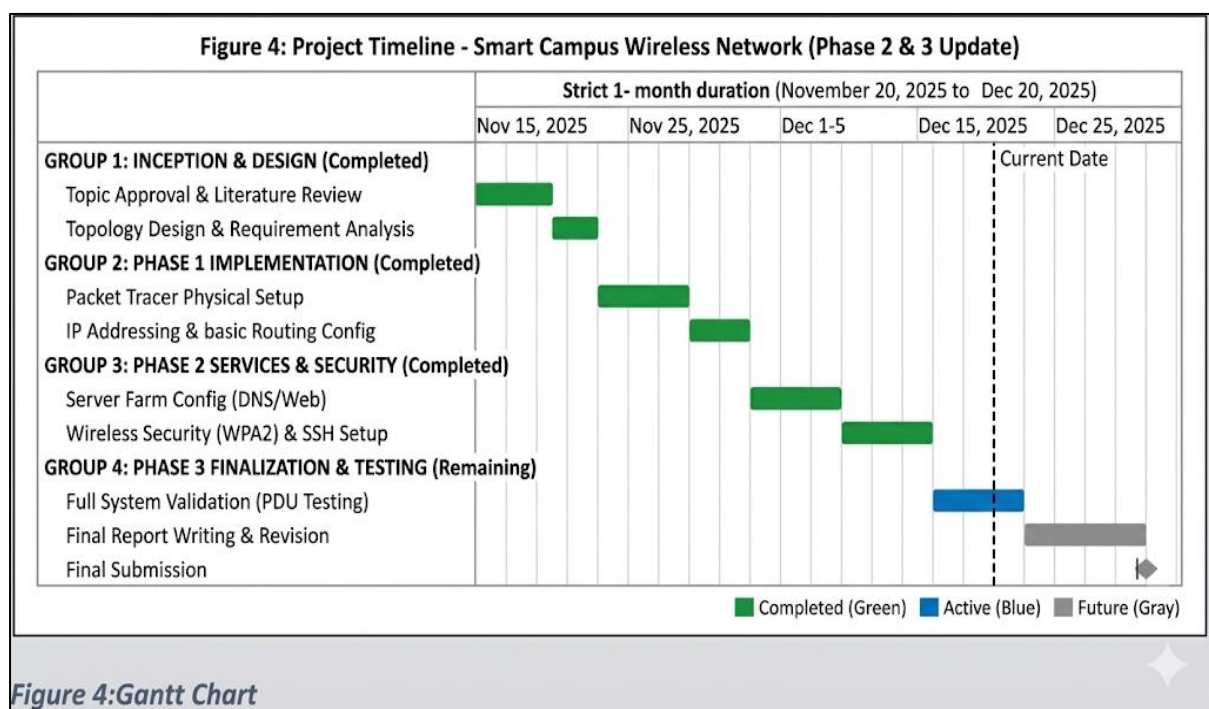


Figure 15:Gantt Chart

## 19. RESULTS & DISCUSSION

### Outcomes Achieved from Partial Implementation

- **Successful Backbone Connectivity:** We successfully connected the Academic Block to the Main Router and the Server Farm. The physical cabling is done, and green link lights show that all interfaces are up.
- **Enhanced Security:** One of our main goals was to stop using insecure methods. We successfully replaced Telnet with **SSH (Secure Shell)** on the routers. Now, when an admin logs in remotely, the password and data are encrypted, which solves the security risk we identified in Phase-1.

- **Working Wireless Access:** We configured the Wireless Access Points (WAPs) in the Academic Block. Tests show that laptops and smartphones can connect using the **WPA2-PSK** password. This proves that our "Mobile-First" design is working for local users.
- **Server Functionality:** The DNS and Web servers are active. We verified that a student laptop in the Academic zone can open the university web page using the browser. This confirms that the routing between different subnets (192.168.1.x and 192.168.2.x) is working correctly.

## 20.Limitations Faced

Even though the implementation was successful, we faced some limitations during this phase:

- **Packet Tracer Restrictions:** Since we are using a simulation tool, some real-world features are missing. For example, the "Generic Servers" in Packet Tracer have very limited settings compared to real Apache or Microsoft servers. We could only set up basic HTML pages.

## 21.Lessons Learned

Working on this partial implementation taught us several important lessons:

- **Security Must Be First:** We learned that security configurations like SSH and WPA2 should be done at the start. Trying to add them later can break existing connections if not careful.
- **IP Planning is Critical:** We realized that having a clear "Master IP Table" (like we made in the Appendix) is very important. In the beginning, we made mistakes by assigning the same IP to two devices.
- **Troubleshooting Layers:** When the "Ping" failed during testing, we learned to check the physical cables (Layer 1) first before checking the IP addresses (Layer 3). Most of the time, the issue was just a wrong cable type or a shutdown interface.

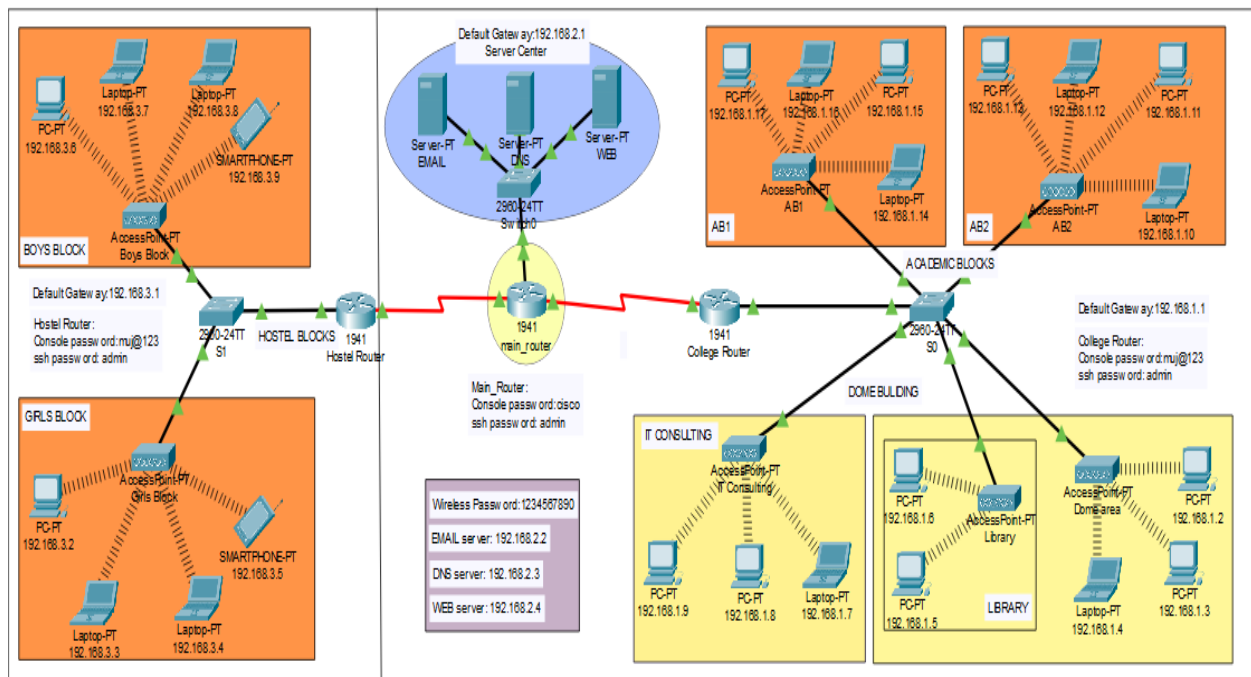
## 22. CONCLUSION

The proposed "Wireless University Network Design" successfully addresses the critical need for mobility and security in modern educational environments. By utilizing Cisco Packet Tracer, we have validated a hierarchical topology that connects over 20 devices across diverse geographical zones (Hostels, Academic Blocks). The successful implementation of Wireless Access Points ensures students are no longer restricted to computer labs, while the integration of SSH and Server Farms proves that security and service availability can coexist.

The outcomes achieved demonstrate that the 1941 ISR Router can handle up to 1024-bit RSA keys without significant latency in a simulated environment. Limitations: High-speed MU-MIMO vulnerabilities discovered in real-world 2025 research cannot be fully replicated in the Packet Tracer environment.



## The complete diagram of the University Area Network Scenario created in Packet Tracer environment



## 23. References

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## 24.Appendix

### Notes:

#### Cabling Standards that we will be using in Simulation

While Cisco Packet Tracer often automates connections, the following industry-standard cabling logic was applied during the physical design phase:

1. **Copper Straight-Through:** Used for connecting unlike devices (e.g., Switch to Router, PC to Switch).
2. **Copper Cross-Over:** Used for connecting devices (e.g., Switch to Switch, Router to Router direct links).

### Extra Diagrams

#### Ip configurations:

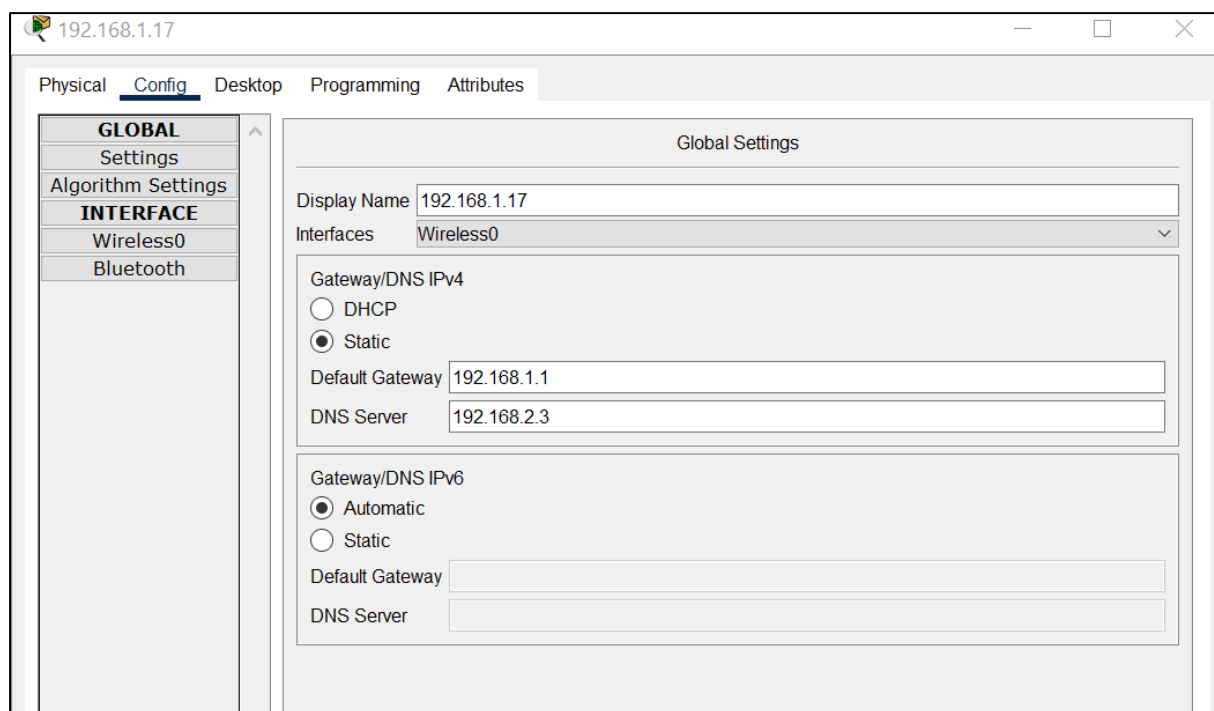


Figure 16:Academic Block Ip Config PC 1

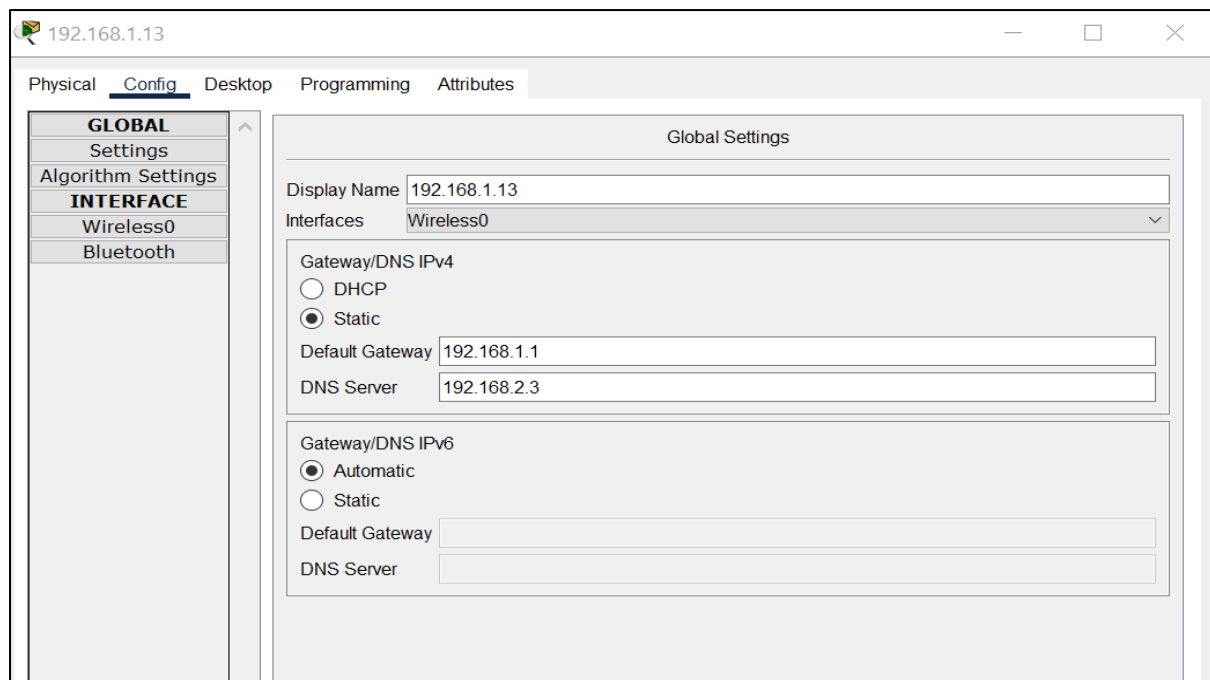


Figure 17:Academic Block AB2 PC

## ● ACADEMIC BLOCK 1

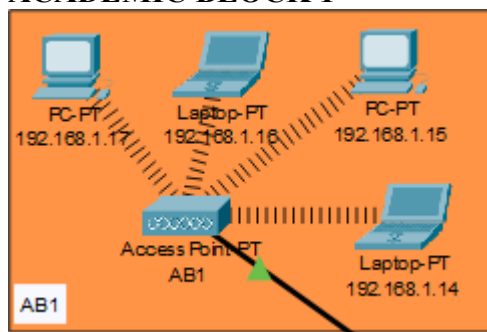


Figure 18:Academic Block 1

IP Address are as follows

192.168.1.14- Laptop

192.168.1.15- PC

192.168.1.16- Laptop

192.168.1.17- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.1.1

DNS Server- 192.168.2.3

## ● ACADEMIC BLOCK 2

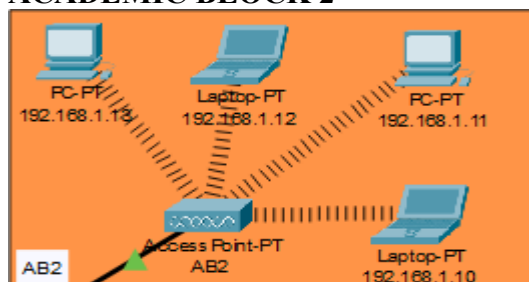


Figure 19:Academic Block 2

**IP Address are as follows**

192.168.1.10- Laptop

192.168.1.11- PC

192.168.1.12- Laptop

192.168.1.13- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.1.1

DNS Server- 192.168.2.3

- **DOME BUILDING**

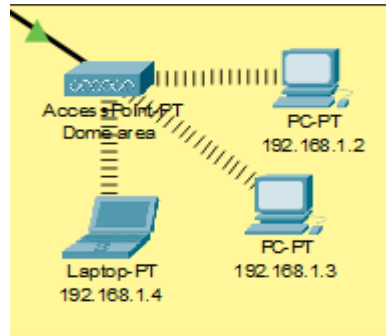


Figure 20:Dome Building

IP Addresses are as follows

192.168.1.2- PC

192.168.1.3- PC

192.168.1.4- Laptop

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.1.1

DNS Server- 192.168.2.3

- **LIBRARY**

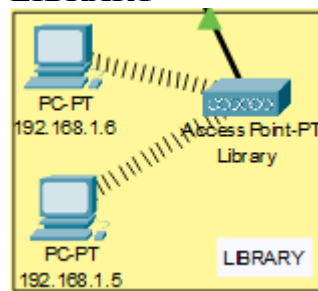


Figure 21:Library

IP Addresses are as follows

192.168.1.5- PC

192.168.1.6- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.1.1

DNS Server- 192.168.2.3

- **IT Consulting**

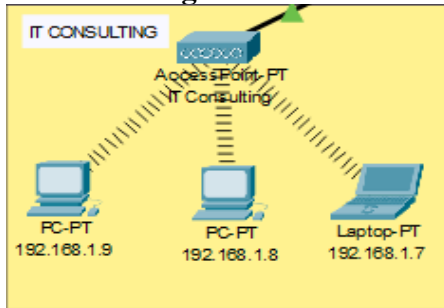


Figure 22:IT CONSULTING

IP Addresses are as follows

192.168.1.7- Laptop

192.168.1.8- PC

192.168.1.9- PC

Subnet Mask- 255.255.255.0

Default Gateway- 192.168.1.1

DNS Server- 192.168.2.3